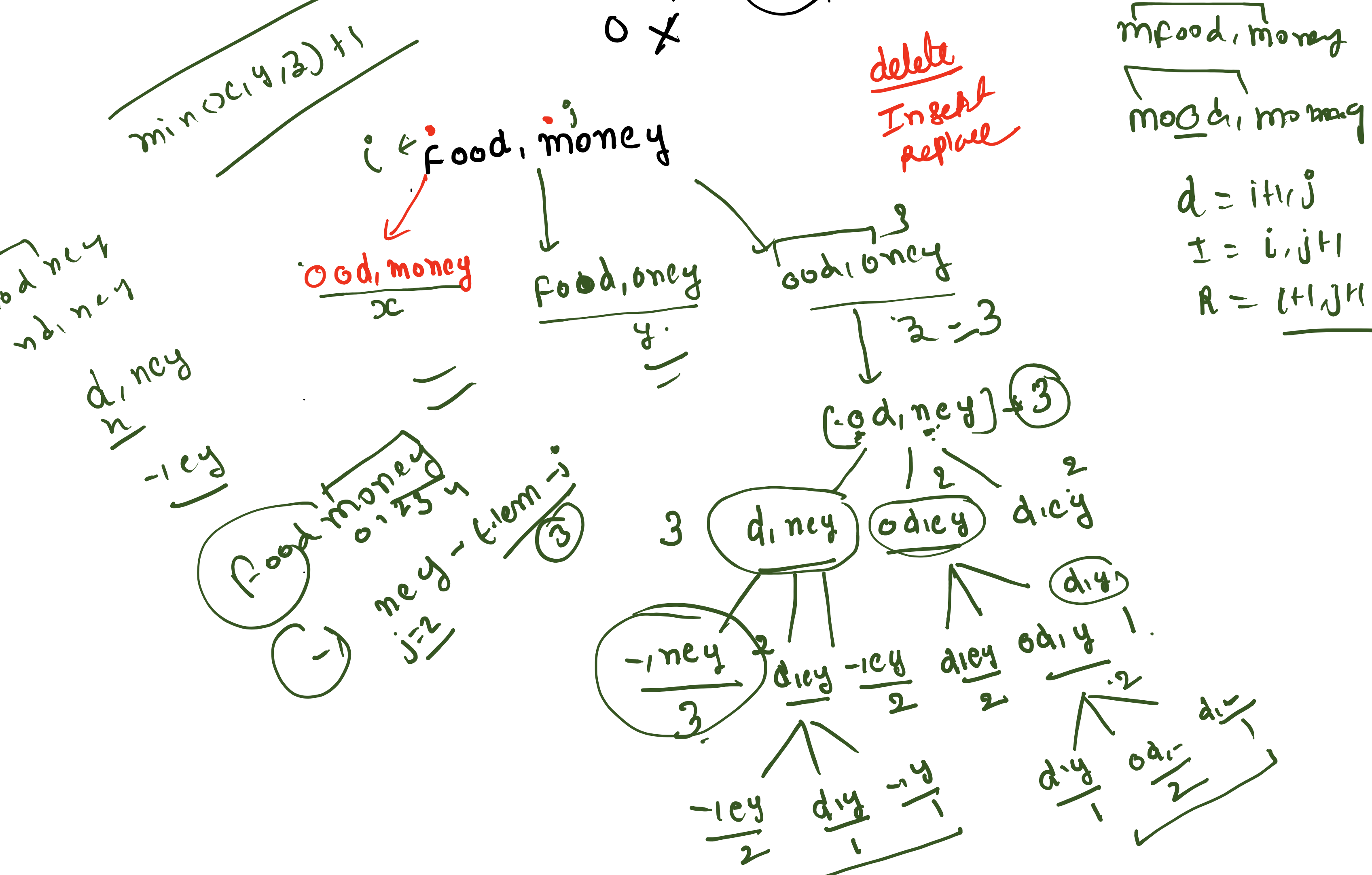
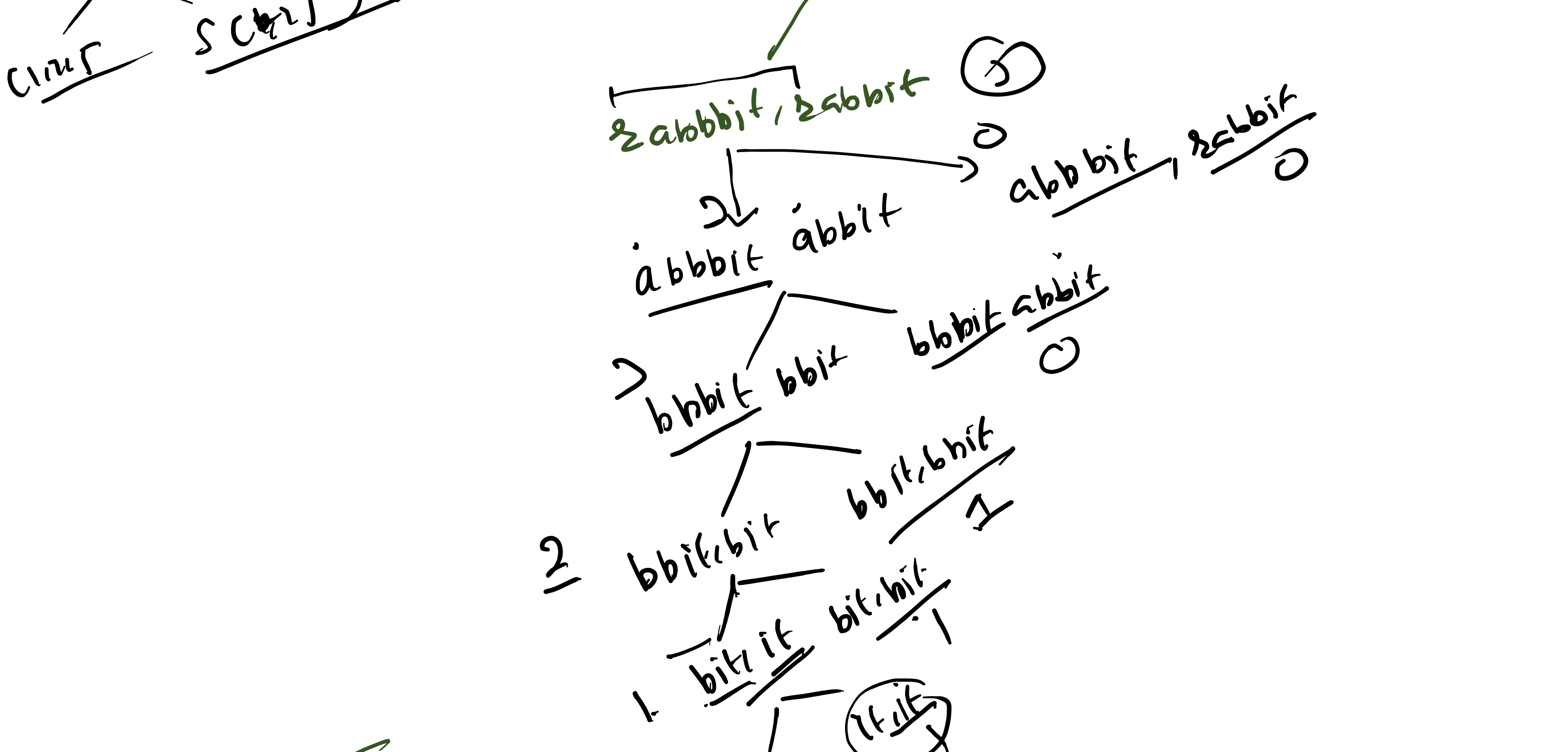
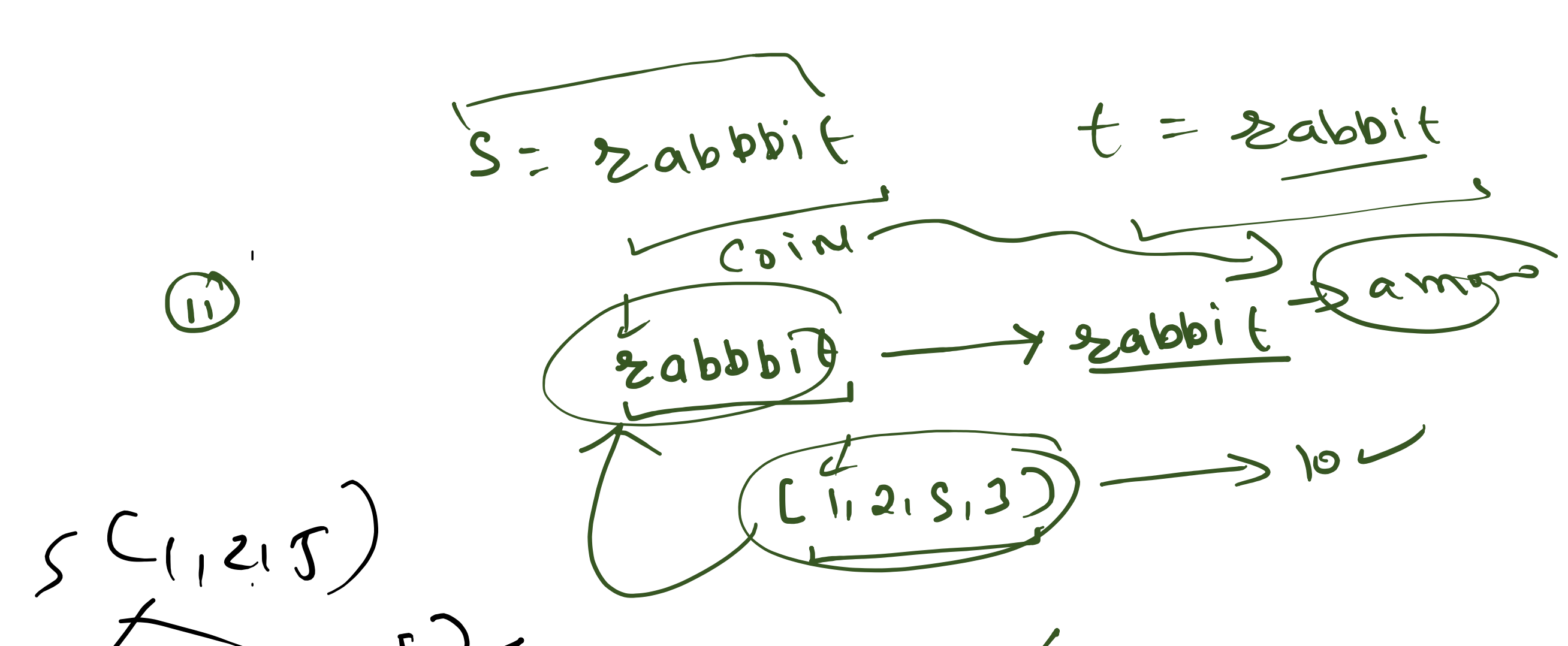
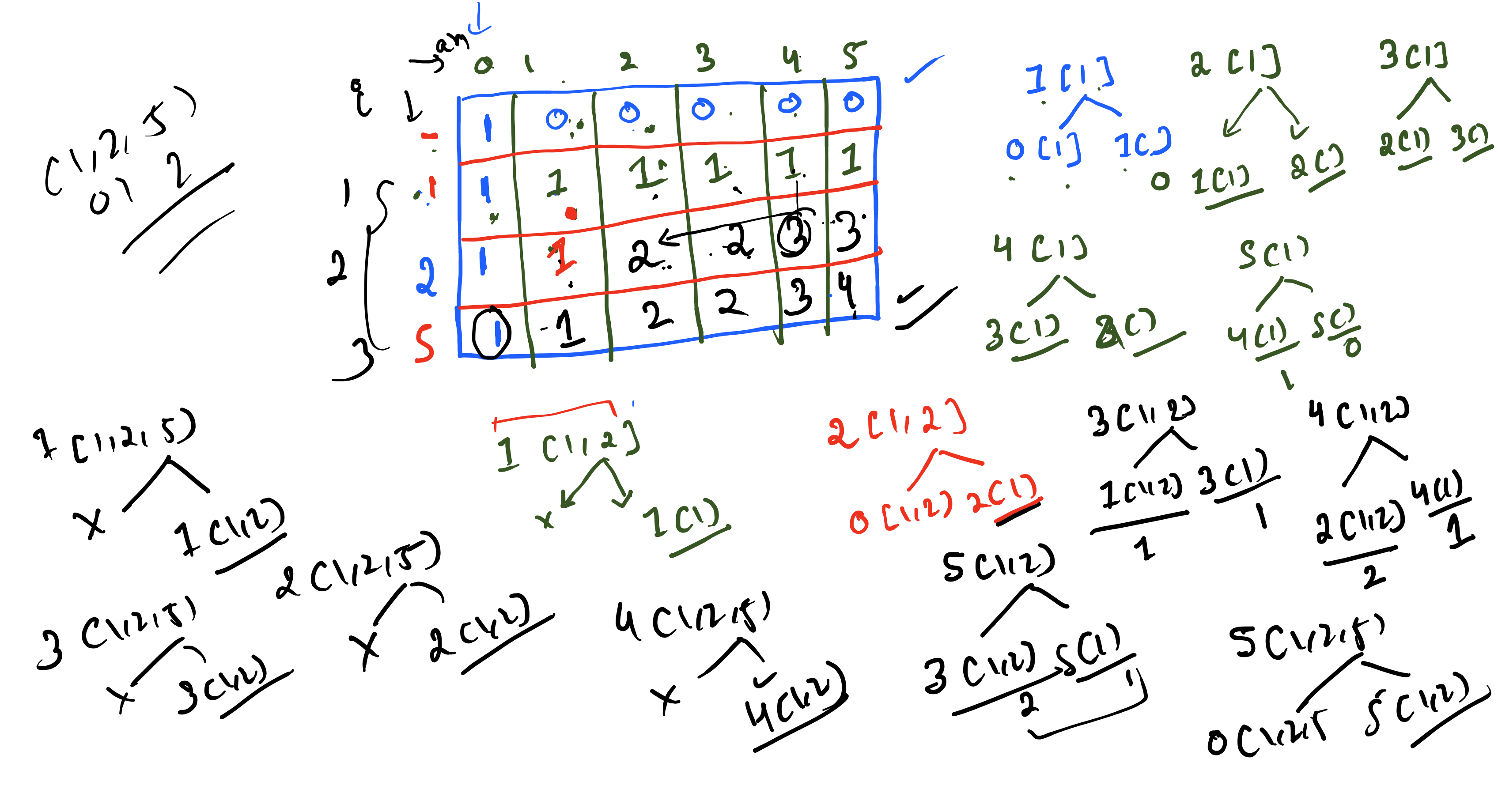
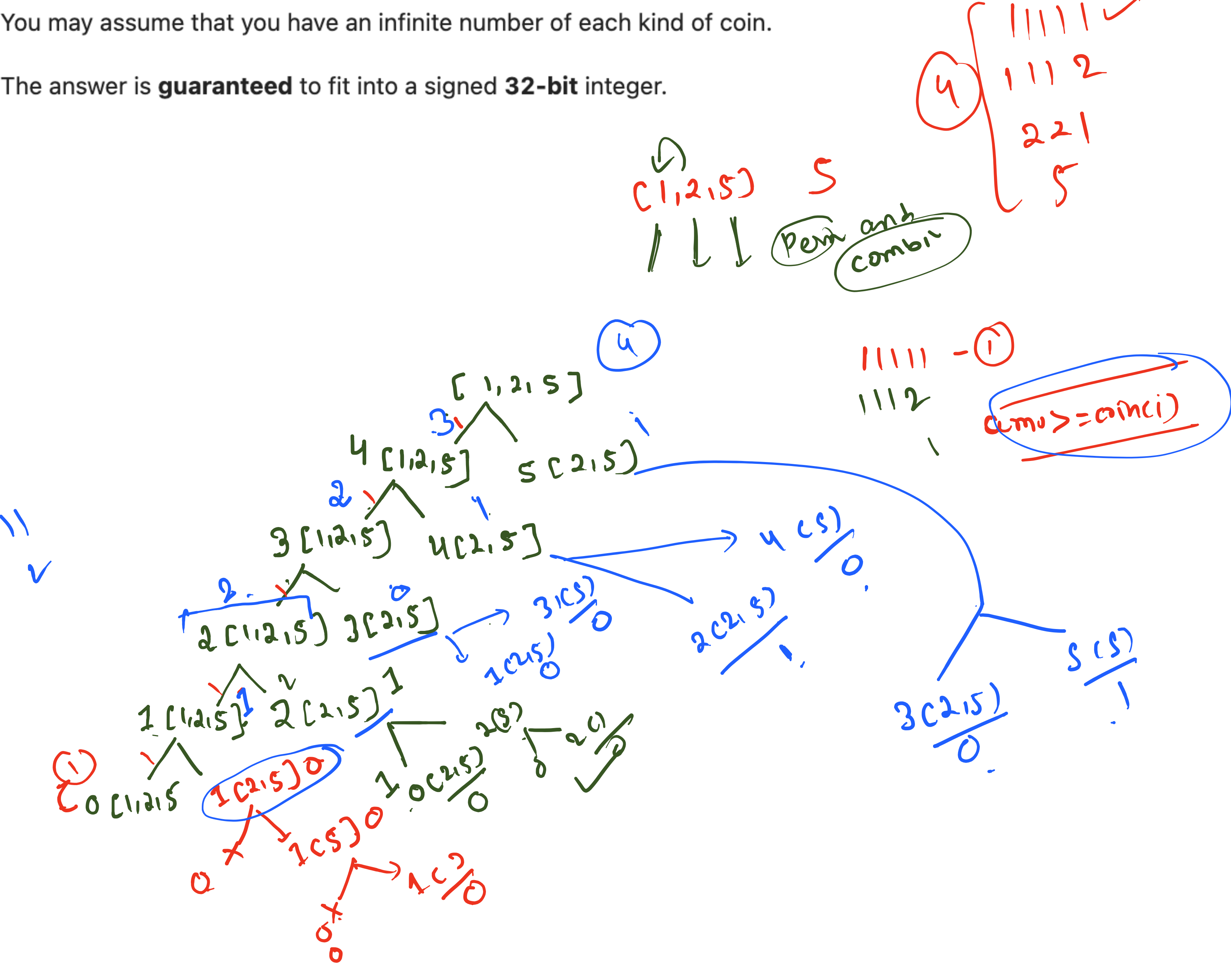


You are given an integer array `coins` representing coins of different denominations and an integer `amount` representing a total amount of money.

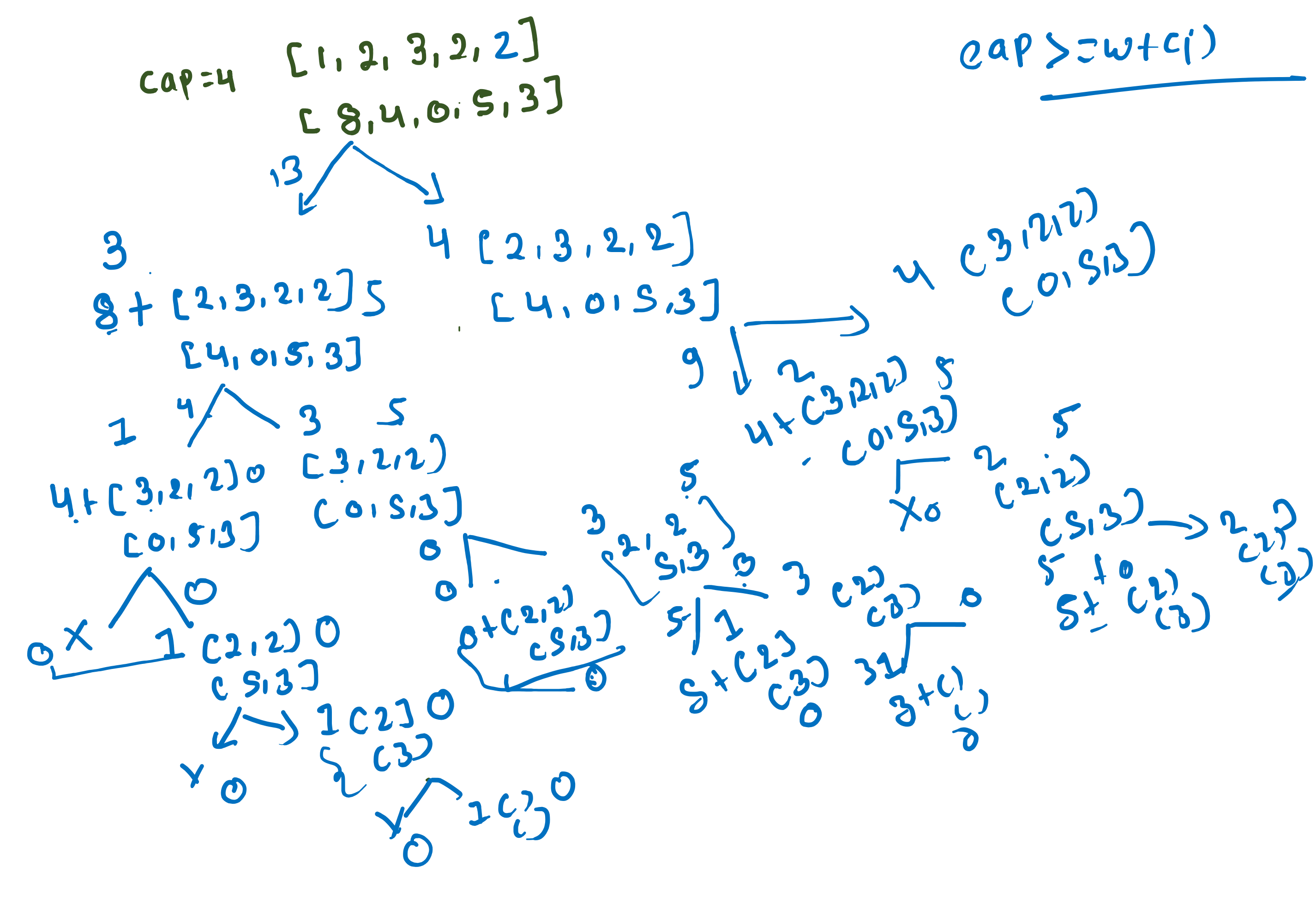
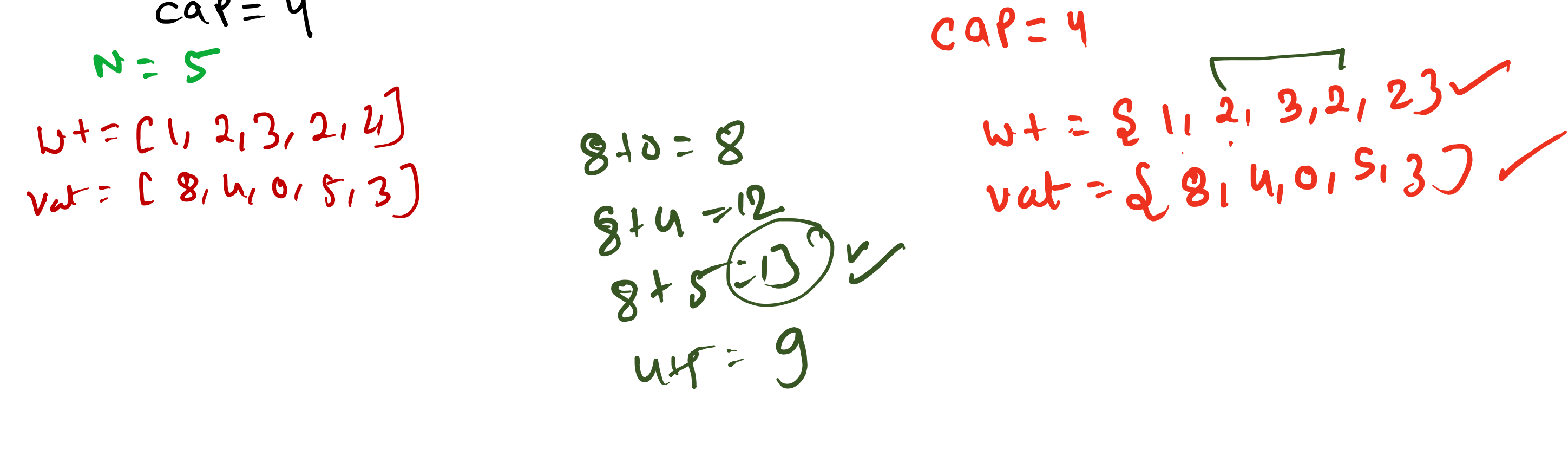
Return the number of combinations that make up that amount. If that amount of money cannot be made up by any combination of the coins, return 0.

You may assume that you have an infinite number of each kind of coin.

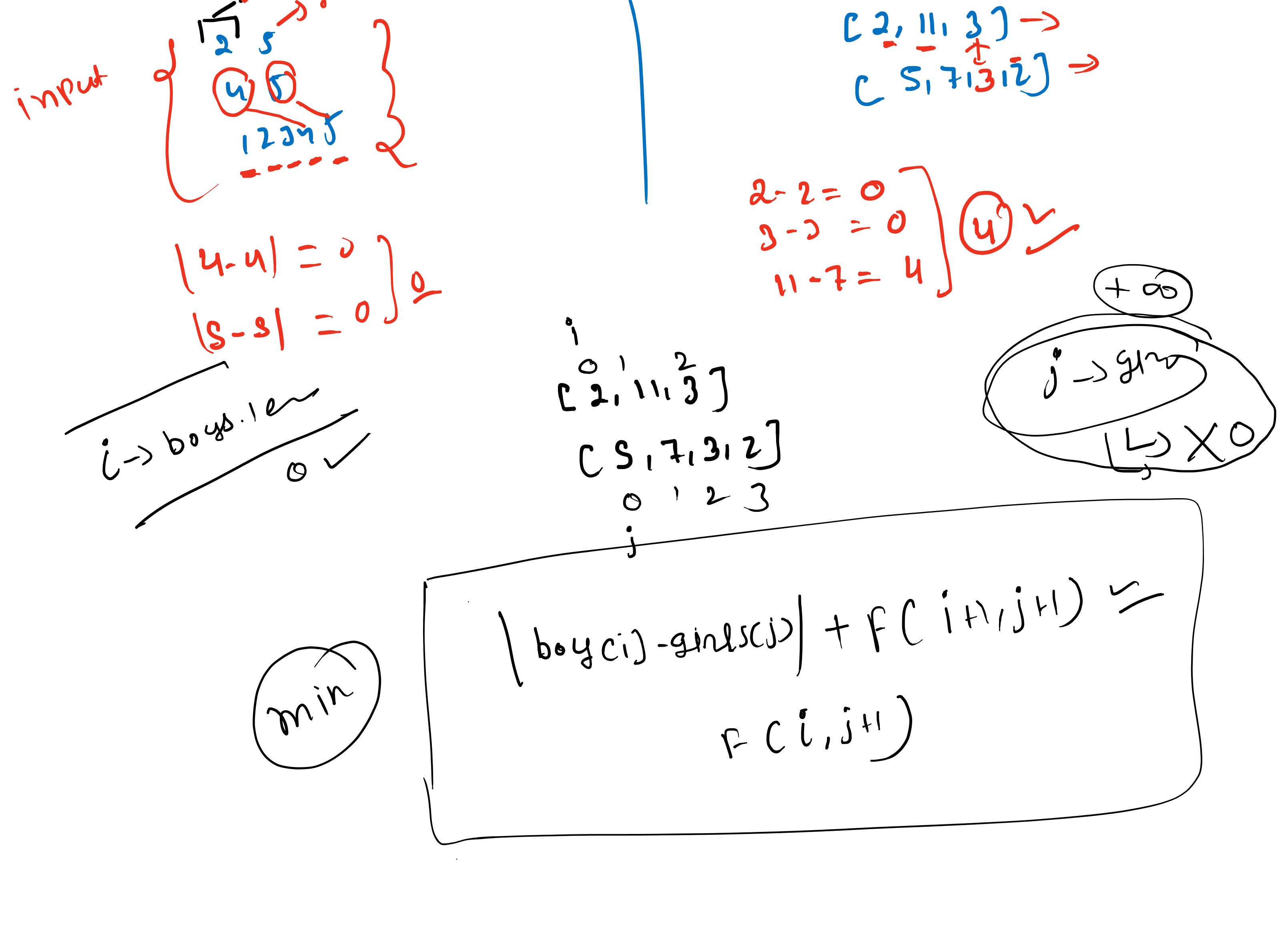
The answer is **guaranteed** to fit into a signed 32-bit integer.



You are packing for a vacation on the sea side and you are going to carry only one bag with capacity S ($1 \leq S \leq 1000$). You also have N ($1 \leq N \leq 1000$) items that you might want to take with you to the sea side. Unfortunately you can not fit all of them in the knapsack so you will have to choose. For each item you are given its size and its value. You want to maximize the total value of all the items you are going to bring. What is this maximum total value?



Valentine Magic
It's Valentine's Day in Russia today, as we all know number of girls in Russia is more than number of boys. hence boys have an extra advantage while choosing girl for the valentine evening. Each boy has certain number of chocolates and each girl has certain number of candies. Now you being the anchor of evening wants to pair all the boys with girls such that the sum of absolute difference between boy's chocolate and girl's candy in a pair is minimized. Ofcourse some of the girls will remain unpaired but that's okay as we are in Russia.



```
public static int Min_Diff(int[] boys, int[] girls, int i, int j) {
    if (i == boys.length) {
        return 0;
    }
    if (j == girls.length) {
        return 9999999;
    }
    int sel = Math.abs(boys[i] - girls[j]) + Min_Diff(boys, girls, i + 1, j + 1);
    int rej = Min_Diff(boys, girls, i, j + 1);
    return Math.min(sel, rej);
}
```

